

“Slide Sense: An Intelligent Landslide Detection and Alert System.”

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Abstract: *Landslides are particularly destructive in mountainous and hilly terrains. When they occur, they cause loss of life, damage to valuable properties, and disruption of essential services. Developed to minimize these effects, Slide Sense is a real-time and cost-friendly IoT-based landslide detection and alert system. It combines seven environmental sensors and a Raspberry Pi Pico W microcontroller, and sends information to the Thing-Speak cloud for visualization as per the locations and conditions of the environment. It is also equipped with a web-based interface designed for situational awareness and emergency management. It provides users with real-time dashboards, risk meters, emergency alerts, and preparatory safety resources. The current work describes the system architecture and the integrated sensors, cloud computing, and web collaboration, illustrating that Slide Sense can work efficiently and smartly and can be easily adapted to other landslide risk zones.*

Keywords: *Landslide, Early warning system, IoT sensors, Real-time monitoring, GIS Mapping, Thing-Speak, Low cost, Risk Meters, Emergency Alerts,*

I. INTRODUCTION

Landslides are one of the most common and catastrophic natural hazards occurring in mountainous and hilly terrains around the world. In India, in a number of states including Uttarakhand, Himachal Pradesh, Kerala, etc., landslides continue to be a recurring threat to life, property, and economy. Yet, heightened as these concerns have become, especially for at-risk populations, a number of them currently continue to monitor their dengue or water-borne diseases the old-fashioned way using human eyes, manual methods, or outdated technology with no alerting and prognostic features. This lack of early warning infrastructure underscores the critical need for intelligent, inexpensive, and scalable interventions deployable in remote, resource-limited communities. Slide Sense is an IoT-powered landslide early warning system that combines actual live environment monitoring, cloud analytics, and public-facing digital

services to solve this issue. The whole system is based on a Raspberry Pi microcontroller, which happens to also sports an onboard Wi-Fi. It connects with seven sensors (three analog raindrop, gas, and soil moisture; four digital ultrasonic, DHT11, vibration, and accelerometer) that are used to collect the information needed to continuously monitor the main parameters for stability of slopes.

The sensors record precipitation, ground movement, soil saturation, temperature and gas release all of which are vital early-warning signs that a landslide could occur. Sensor readings are sent to the Thing-Speak cloud and visualized using dynamic graphs. In order to both democratize and personally use this data, a responsive website is set up as the frontend for the system. Website includes a dynamic risk meter, sensor values in real-time on a dashboard with alerts and Thing-Speak charts. It also includes GPS-based location tracking, emergency contact cards, and a one-

click SOS feature to notify authorities in case of imminent danger. Additionally, the site provides preparatory aids such as safety guidelines, evacuation tips, and educational resources to help users respond effectively during a landsliding.

In addition to having a functioning frontend and cloud integration, the automated alerts, user authentication, emergency coordination, dynamic data retrieval, and backend dynamic data retrieval functions have been automated. It establishes the secure connections necessary for real-time interactivity and unobstructed communicative exchanges between the microcontroller and the web interfaces, cloud, and backend. Slide Sense is a Modular, scalable, and customizable to a wide variety of landscapes. Slide Sense is an adaptable candidate for proactive disaster management in areas where landslides pose a risk. Transforming the risk of landslides on a site, Slide Sense provides risk monitoring, proactive risk communication, and risk mitigation.

II. RELEVANT BACKGROUND

In India's hilly and mountainous regions, landslides become a significant threat to life, property, and nature. Uttarkashi and Wayanad, Nainital, and Wayanad have been hit by serious, repeated landslides. Key trigger factors are intense rainfall, soil erosion, earthquakes, and human actions such as deforestation and reckless construction. The affected areas' reliance on old and manual monitoring systems, which are not real time responsive and predictive, has created failure to respond to systems and resulted in late warnings. As the frequency and severity of these events increase, the lack of preparedness and response systems increases the susceptibility of the affected population.

Recent developments in geospatial analysis, remote sensing, and machine learning to improved landslide susceptibility assessment and risk prediction. Unfortunately, many of these techniques are not viable in practice, whether because they are expensive to deploy, present complicated challenges for real-world deployment, or simply do not scale. Research has proved that the IoT solutions, including the sensor network and cloud integration into the system, can provide real-time environment monitoring and early-warning [14]. As an example, fuzzy logic controllers,

supervised learning algorithms, and PS-InSAR techniques have already been used with success for the detection of slope instability and ground deformation. The problem however is achieving integration of these technologies into a single community accessible platform.

Slide Sense fills this gap by enabling a complete early warning landslide solution, including inexpensive hardware, cloud-based analytics and an interactive web UI. Housed on a Raspberry Pi Pico W microcontroller, it combines seven sensors, three analog (raindrop, gas, soil moisture) and four digital (ultrasonic, DHT11, vibration), to monitor the most important environmental parameters. Expectation data are fed to the Thing-Speak cloud for visualization and analysis, a dedicated website is created to render real-time dashboards, risk meters, emergency alerts and preparatory aids (safety tips, evacuation resources).

The significance of Slide Sense is in its potential to democratize landslide monitoring by making high-speed intelligent detection system available for risk-affected communities. It fills the void between academic research and field deployment, making feasible a scalable solution that applies to different types of landscapes as well as disaster-prone areas. Slide Sense advances a proactive paradigm of disaster risk reduction and community resilience by combining sensor data, cloud computing infrastructure, with public engagement technologies.

III. LITERATURE REVIEW

Given the rise in both frequency and intensity of slope failures in the country, landslide detection and prediction in the hilly terrains of India has become an active area of research. Several works have addressed the prediction of landslide occurrences using sensor-based systems, the implementation of machine learning algorithms, and remote sensing technologies in the Susceptibility mapping. Gopi et al. [12] have developed a cloud-based landslide detection technology using innovative IoT devices like soil moisture sensors, GPS devices, and soil moisture sensors, and have integrated the system with soil moisture sensors and accelerometers. The provided MQTT soil moisture sensors and accelerometers soil moisture sensors framework has provided an

automated solution with landslide zone real time alerts in the control center. In a similar study, Kainthura and Sharma [9] have provided a supervised learning study on Varunavat Parvat, Uttarkashi, by forecasting landslide susceptibilities using Support Vector Machine, ANN, and Logistic Regression techniques. The study has found slope instability and Rainfall as the primary triggering factors, while SVM and Logistic Regression were found to have the highest predictive landslide forecast capabilities. In the study of Bhangale et al. [5], a fuzzy logic-based landslide detection system is presented in which the proposed method integrated sensors for soil moisture, vibration, rain, and an accelerometer. Thresholds were classified as No Alert, Alert, and High Alert, accordingly, using ThingSpeak cloud and MATLAB analytics, thus proposing a suitable real-time monitoring system. Venkatachalam et al. [4] modeled a rainfall-induced landslide using DEM and GIS to predict runout patterns using fractal analysis. In their work, they provided a robust framework for evaluating the risks and planning the mitigation in the Western Ghats. Aggarwal et al. [10] proposed a landslide monitoring system based on the Raspberry Pi using video stream analysis and tracking of the centroid. The low-cost setup enabled real-time detection and provided alerting through an Android application, showing the potential of computer vision methods in geohazard monitoring. Ado and Amita [15], on the other hand, prepared the landslide susceptibility maps of Meghalaya using SVM, trained on 1005 landslide points and 19 causative factors. Their model achieved an AUC of 0.91 with feature selection and presented road proximity, slope, and soil moisture as the main predictors. Achu et al. [13] conducted a study on the comparison of DNN and RF models for landslide susceptibility in Wayanad, Kerala, where DNN surpassed RF with an AUC of 0.934, with slope angle, convergence index, and road distance being the three most influential variables. In Puthumala, Kerala, Jacob et al. [1] employed Sentinel-1 data for PS-InSAR and surface deformation mapping. Interferometric results were validated with post-landslide LULC, NDVI, and field surveys showing millimeter accuracy in terrain shift detection. In Nainital, Uttarakhand, Dwivedi et al. [2] for the monitoring of landslide used StaMPS PS-InSAR technique on ENVISAT data. This revealed critical displacement zones of 17 mm/year which were

validated with the hazard zonation maps. In another study, Varangaonkar and Rode [10] conducted a comparative study of SVM and ANN for landslide prediction using remote sensing and GIS data. In this study, the ANN achieved an accuracy of 93.38%, while SVM lagged behind at 91.78%. In addition, their real-time prediction system utilized NDVI combined with other robust features for improved classification. Singha and others [14] designed a sensor-based landslide detection system for the southeastern part of India. Their prototype built with the Arduino Uno microcontroller and an MPU6050 accelerometer and soil moisture sensors, detected slope alteration and moisture content changes. The prototype achieved real-time performance and system responsiveness at varying climates, thus widening its low-cost system potential in economically sensitive environments hyper-sensitive to such deployments. These studies highlight the growing need for geospatial analysis, sensor networks, and machine learning in landslide prediction and monitoring. While the use of GIS and remote sensing approaches constitutes the core upland monitoring, the use of sensor systems, real information and smart systems for decision making, offers precise and timely upland early warning

IV. SYSTEM OVERVIEW

A. Objectives

- Create an affordable and easy-to-scale landslide detection system utilizing IoT and cloud tech.
- Track essential environmental variables as they change with our 7 real-time integrated sensors.
- Send sensor data collected in the field to the Thing-Speak cloud for visualization and data analysis.
- Provide a fully responsive web interface for public access to the sensor dashboard and alerts.
- Automated logic for alerts and emergency notification tools gives the system the ability to warn users in advance.
- Community emergency preparedness is supported with resource materials.

- System architecture is designed to be modular to suit a specific terrain and deployment situation.
- Secure data handling and risk classification is offered by implemented backend services.

B. Key Features

- Continuous Monitoring: Continuous observation of rainfall, soil moisture, vibration, gas concentration, temperature, humidity, displacement, and slope movement.
- Cloud and Mobile Analytics: Performing visualization and historical trending analysis on Thing-Speak & additional insights on MATLAB.
- Web Interface Live Monitor System Live dashboard with an active risk meter, alert banners, and integrated charts.
- Arbitrary Class Risk Levels Automated risk tagging (Safe, Warning, Danger) with visual notifications and alerts.
- Real-Time Location Monitor System 1-SOS Dispatch, 3 geolocation, and 3 nearby safe zone recognition.
- Emergency SOS System One-click alert to authorities with user location and sensor status for rapid response.
- Website-Integrated Safety Aids Evacuation tips, emergency phone numbers, warning signs, and guidelines.

V. HARDWARE ARCHITECTURE

The design of Slide Sense's hardware incorporates inexpensive, the design of Slide Sense's hardware incorporates low-cost, energy-efficient, and modular components for uninterrupted ecological monitoring in regions susceptible to landslides. The hardware framework consists of various components, with the Raspberry Pi Pico W being the most important. The Pico W Raspberry Pi Pico W serves as the system's CPU and core Sensor- microcontroller. It also serves as the core microcontroller because of its miniature size, low power consumption, and built-in Wi-Fi. It serves as the central processing unit interfacing with a total of seven sensors- three analog and four digital that monitor key parameters influencing slope stability. The analog sensors include a Raindrop sensor

to detect precipitation intensity, a Gas sensor (MQ2) to emit smoke in case of fire and a Soil Moisture sensor to assess water saturation levels. These sensors are connected to the Raspberry Pi Pico W's ADC pins for analog-to-digital conversions. The digital sensors include an Ultrasonic sensor for measuring ground displacement, a DHT11 sensor for temperature and humidity, a Vibration sensor for seismic activity and an Accelerometer for detecting slope movement. These sensors are interfaced via GPIO, I2C or SPI protocols depending on the sensor type.

Micro Python is used to program the microcontroller which helps collect sensor readings, preprocess data, and send it to the Thing-Speak cloud platform via wireless HTTP requests. It is powered through a regulated 5V source. The system is meant to run continuously with little maintenance. The design performs real-time data acquisition, transmits data to the cloud, and provides local notifications using a buzzer or display modules. This suit of hardware facilitates dependable and programmable use in the middle of nowhere, thereby forming the basis of Slide Sense's smart landslide detection.

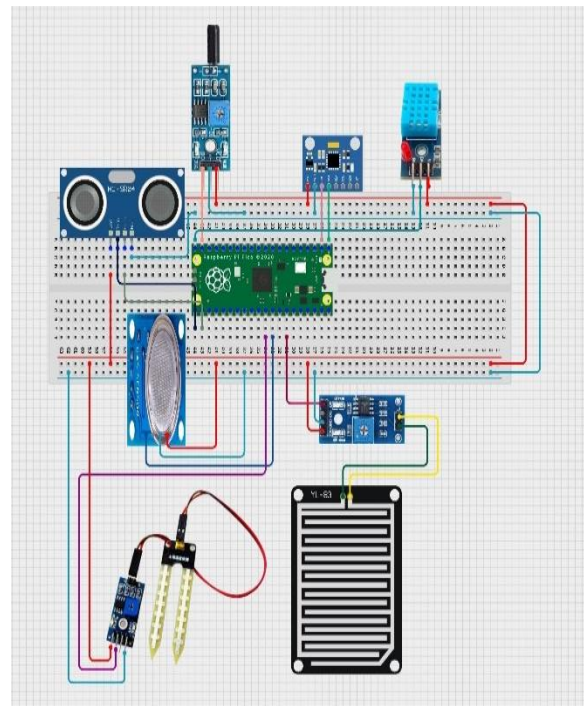


Figure 1 Circuit Diagram of Slide Sense Hardware

Sensor Type	Sensor Name	Interface Type
Analog	Raindrop Sensor	ADC
Analog	Gas Sensor	ADC
Analog	Soil Moisture	ADC
Digital	Ultrasonic Sensor	GPIO (Trigger/Echo)
Digital	DHT11	GPIO
Digital	Vibration Sensor	GPIO
Digital	Accelerometer	GPIO (I2C/SPI)

Table 1. Sensors Used in Slide Sense

VI. SOFTWARE AND CLOUD INTEGRATION

A. Data Transmission

The first layer of data transmission is controlled by the Raspberry Pi Pico W microcontroller. It collects information from seven environmental sensors. Gas, soil moisture and raindrop sensors are analog sensors; whereas ultrasonic, DHT11 and vibration sensors are digital types; the accelerometer sensor is also a digital one. For programming, Micro Python is used for routine sampling from the sensor, data preprocessing, and formatting for transmitting to the cloud. The processed data from the Pico W is sent to the ThingSpeak cloud via HTTP POST requests using its integral Wi-Fi module. In all of these uploaded data we put value with the time stamp and processed fields individually in the ThingSpeak channel so that transmission can be done proper and reliable way.

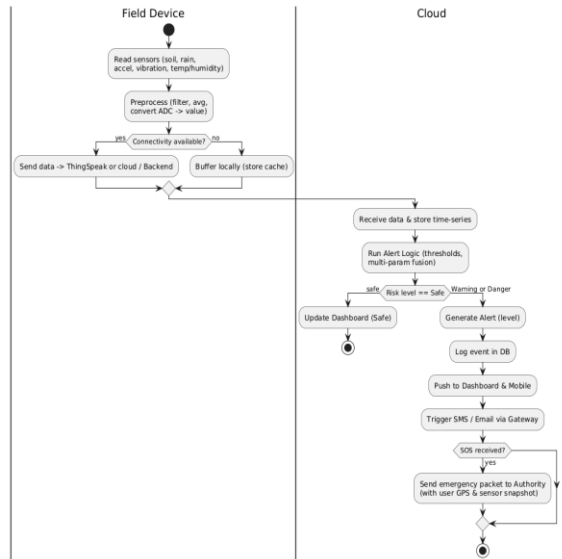


Figure 2 Flow of data

B. Thing-Speak Cloud

Data are stored in, visualized in, and analyzed with Thing-Speak as the central cloud application. All sensor parameters are stored in a separate field in a Thing-Speak channel for easy data management. Real-time graphing of data allows users to keep track of environmental trends as they change. Historical data is preserved and may be retrieved for analysis. Thing-Speak allows you to do MATLAB analytics in the cloud, which can be used for anomaly detection, computing moving averages, and training predictive models. This cloud connectivity also means sensor data is not only available remotely, but it's now intelligently curated for analysis and decision-making in real-time.



Figure 3 ThingSpeak Cloud

C. Alert Logic

The alert mechanism has been designed to determine risk levels and issue alerts based on defined thresholds in sensor data. The model evaluates various combinations of soil moisture, rainfall, vibrations, and slope movement and provides the basis for assigning one of the three risk levels, then termed 'Safe', 'Warning', and 'Danger'. When the determined thresholds are passed, the backend system triggers alerting actions designed for dashboards and includes visual banners, activation of the risk meter, and notifications by email and SMS (planned for the future). This entailed logic that provides alerts that are

contextually relevant and timely. Thus, potential preventive actions can be undertaken.

VII. WEB INTERFACE

The Slide Sense web interface presents time-sensitive danger notifications and emergency services to end-users and officials. This interface has been made to operate on multiple devices.

Real-time Dashboard: A primary feature of this dashboard is that it displays the unit's sensor data readings in real-time. It closes the gap on transparency concerning the data that dictates the state of the environment and the likely (un)likely slope of the terrain. These data points include moisture level in soil, precipitation, and geomorphology change.

Dynamic Risk Meter: Effective communication of risk status is essential. This alters the user's risk status of (landslide) at the moment and changes as the backend systems risk level changes in real-time (automated level changes of Safe, Warning, or Danger).

Alert Display: The dashboard displays warning signals for imminent danger. to all active end-users when the alert level changes to Warning and Danger.

ThingSpeak Chart Integration: The interface includes time-related sensor data so that the end-users see active (now) readings and the data history.

Emergency SOS System: A distinctive and important interface feature is the SOS push button. This feature allows the user to (automated) send the current sensor readings and the user's geolocation to officials.

VIII.IMPLEMENTATION WORKFLOW

The functioning installation of the Slide Sense system uses particular multi-stage flows, merging the physical system with the graphic web interface.

Acquisition of Data (Hardware Layer): Physically, the Pico W microcontroller extracts data from the seven sensors one at a time, polling data iteratively in a while loop for an indefinite time. For interfacing with sensors, data printing, and conversions from analog to digital (Raindrop, Gas, Soil Moisture) sensors, MicroPython is used if relevant at the time of sensing.

Data Transmission (Communication Layer): The Pico W has a microprocessor that organizes the data from the sensors and sends it to the Thing-Speak cloud on its own using HTTPS POST requests with a timestamp and some sensor sample data that has been pre-processed for specific fields at requested time intervals. Data sent using POST is secured. The microcontroller's Wi-Fi feature enables internet access.

The cloud analysis and alerting (Backend Layer): Data sent to Thing-Speak needs to be visualized on the platform. The MATLAB platform analysis can be used for anomaly detection. The Alert Logic module deployed scans sensor read zero data intervals and compares them to threshold values. The module risks classifies on a tiered scale to Safe, Warning, and Danger.

Notification and Visualization (Frontend Layer): Customers can view a web interface that streams data in real-time and shows the current alert level. The system automatically generates changes to alert level on the risk meter, alerts, and components in the data and dashboard visualizations that are under the cloud and backend services.

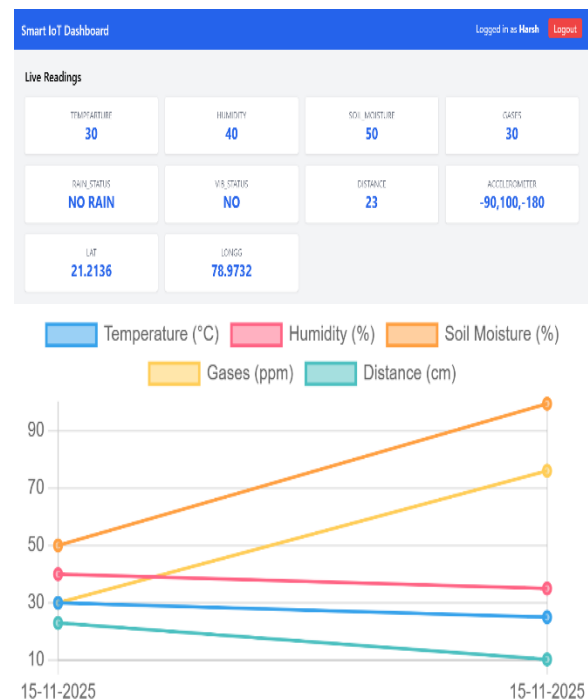


Figure 4 Dashboard Readings

IX. EXPERIMENTAL RESULTS

A. Sensor Readings

Validation of sensor reaction was done through replicating significant triggers of landslides, including the following:

Water Content - The Soil Moisture Sensor was able to identify different levels of water saturation and was able to perform a high saturation reading, which in the system alert logic was able to contribute to an increased level of risk, replicating conditions prior to a landslide.

Ground Movement - The Accelerometer and Vibration Sensors were able to identify and respond to small movements of the slope and to minor levels of seismic activity. We were able to cause a change in the slope angling shift which was able to activate a Reporting level on the risk-measuring device. We were able to confirm the system's capabilities in monitoring the stability of the slope.

B. Cloud Performance

Data Reliability - All of the sensors were operating seamlessly, transmitting their readings successfully for the entirety of the testing periods with a high level of data reliability, creating a complete dataset on Thing-Speak for the tests conducted.

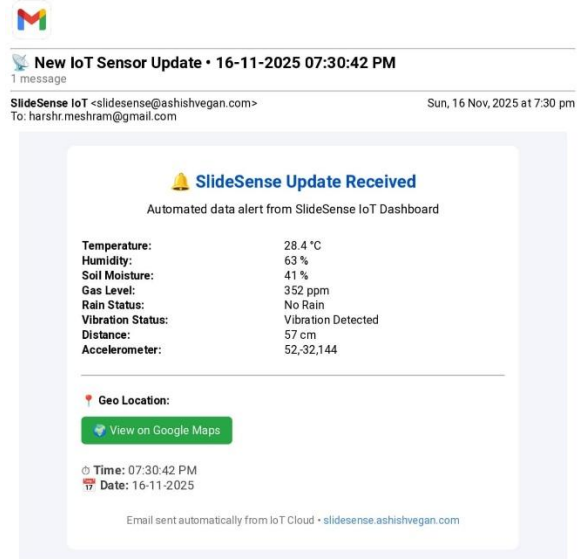
Alert Accuracy - The Alert Logic was able to automatically and in real-time assign the specific scenario to one of the three risk levels - 'Safe', 'Warning', 'Danger', and in doing so confirmed the reliability of risk scenario classification. The environmental data was dynamically displayed in real-time on Thing-Speak which successfully confirmed the real-time environmental data tracking.

C. Website Functionality

The web interface performed as designed: The Live Dashboard displays all sensor values, the Risk Meter, and ThingSpeak charts instantaneously.

The Emergency SOS System successfully relays on the user's location and sensor status to a simulated

authority notification system, confirming the functionality of the one-click alert feature for rapid response.



X.CONCLUSION

Slide Sense is a groundbreaking improvement in the creation of smart, accessible, and expandable systems for the recognition of landslides. The system incorporates 7 low-cost IOT sensors, a Raspberry Pi Pico W microcontroller, and Thing-Speak cloud for the constant collection and monitoring of critical environmental variables affecting slope stability.

The strong configuration permits uninterrupted real-time collection and transfer of information analyzed by the system in a complianced automated alert logic and determines a risk category to issue timely notifications. The system offers fully interactive web interfaces - including real-time displays, a risk meter, emergency notification, and preventative alert resources - to meet the urgency of early warning systems for vulnerable populations.

The system's experimental results demonstrate the system's performance, data integrity, and accuracy of risk categorization in real time. This illustrates that Slide Sense works efficiently and intelligently and can be adapted to other at-risk landslide areas. The project promotes a positive framework for disaster risk mitigation and community resilience.

XI. FUTURE SCOPE

Incorporation of GIS mapping and satellite images will improve Slide Sense real-time hazard display and analysis. Terrain analysis and hazard visualization will be accessible on a larger scale. Secure data sharing through blockchains will be transparent and unalterable and provide a communication channel between authorities and disaster management teams. A mobile smartphone application will give users real time alerts, SOS pressing, GPS and tracking, and SOS access.

Predictive modeling for early landslide forecast and multi-risk detection (flood, earthquake) will be powered by artificial intelligence. Enhanced automation through cloud and scalable architecture will allow Slide Sense to manage large sensor networks and fast reliable disaster warning. With these, Slide Sense will have be provided with a sophisticated, smart, and community-aware early warning system. These will be powered by artificial intelligence

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